

2016.0 RANGE ROVER (LG), 412-00

CLIMATE CONTROL SYSTEM

- GENERAL INFORMATION

SPECIFICATIONS

Climate Control

NOTE:

NAS Vehicles.

TYPE	DESCRIPTION
Heating, ventilation and air conditioning (A/C) unit	Dual zone or four zone automatic system fitted controlled by automatic temperature control module (ATC) module.
A/C compressor	Sanden, internally controlled variable capacity A/C compressor
Refrigerant	HFO-1234yf

Climate Control

NOTE:

EU Vehicles.

TYPE	DESCRIPTION
Heating, ventilation and air conditioning (A/C) unit	Dual zone or four zone automatic system fitted controlled by automatic temperature control module (ATC) module.

A/C compressor	Sanden, internally controlled variable capacity A/C compressor
Refrigerant	HFO-1234yf

Climate Control

NOTE:

ROW Vehicles.

TYPE	DESCRIPTION
Heating, ventilation and air conditioning (A/C) unit	Dual zone or four zone automatic system fitted controlled by automatic temperature control module (ATC) module.
A/C compressor	Sanden, internally controlled variable capacity A/C compressor
Refrigerant	R134a

Lubricants, Fluids, Sealers and Adhesives

CAUTION:

HFO-1234yf systems use the refrigerant oil (Sanden SPA2) and is not compatible with any other refrigerant oils.

NOTE:

The refrigerant oil (Sanden SPA2) is compatible with R134a systems.

ITEM	SPECIFICATION
A/C compressor oil type	Sanden SPA2
A/C compressor oil - vehicles fitted with 2 zone	80 ml

General Specification

ITEM	DESCRIPTION
A/C compressor	Sanden
Sensor locations:	
Ambient temperature sensor	Left door mirror
Cabin temperature sensor	Mounted behind a grill on the drivers side of the instrument panel, adjacent the steering column
Cabin humidity sensor (optional fit)	Located within the interior mirror and comprises 3 individual elements
Evaporative temperature sensor	Mounted on the right side of the heater casing
High pressure sensor	Located between the condenser and the thermostatic expansion valve
Pollution sensor	Mounted on the inboard side of the blower assembly
Solar sensor	Center of dash top

Capacities

ITEM	VEHICLES FITTED WITH 2 ZONE	VEHICLES FITTED WITH 4 ZONE
Total system capacity refrigerant	800 grammes $\pm$ 10 grammes	1150 grammes $\pm$ 10 grammes

Refrigerant oil adding capacities for replacment parts

NOTE:

Rotate the A/C compressor shaft at least 6 to 8 turns when draining the refrigerant oil.

ITEM	MILLILITERS
A/C compressor - vehicles fitted with 4 zone	Add 60 ml.
A/C condenser core and desiccant bag	Add 25ml oil
A/C evaporator	Add 30ml oil
A/C compressor	1. Drain old A/C compressor. With drain plug removed and ports uncapped, rotate shaft to remove A/C compressor oil and measure the amount of oil captured. 2. Drain new A/C compressor into a clean vessel. With drain plug removed and ports uncapped, rotate shaft to remove oil. Then add back a quantity of the new oil that is identical to the quantity of oil removed from the old A/C compressor. However, if this quantity is less than 30ml, then make it up to 30ml.
A/C lines - if air conditioning has been operational.	Add 5ml oil per A/C line
A/C system after flushing - with new compressor installed	No oil should be used- new oil in new compressor is sufficient
A/C system after flushing - without a new compressor installed - remaining A/C compressor oil is to be drained. Compressor and expansion valve must not be flushed (removed from the circuit)	Add 80ml oil
A/C compressor drain plug	Torque 12Nm